

Real-Time Omnidirectional Roaming in Large Scale Indoor Scenes

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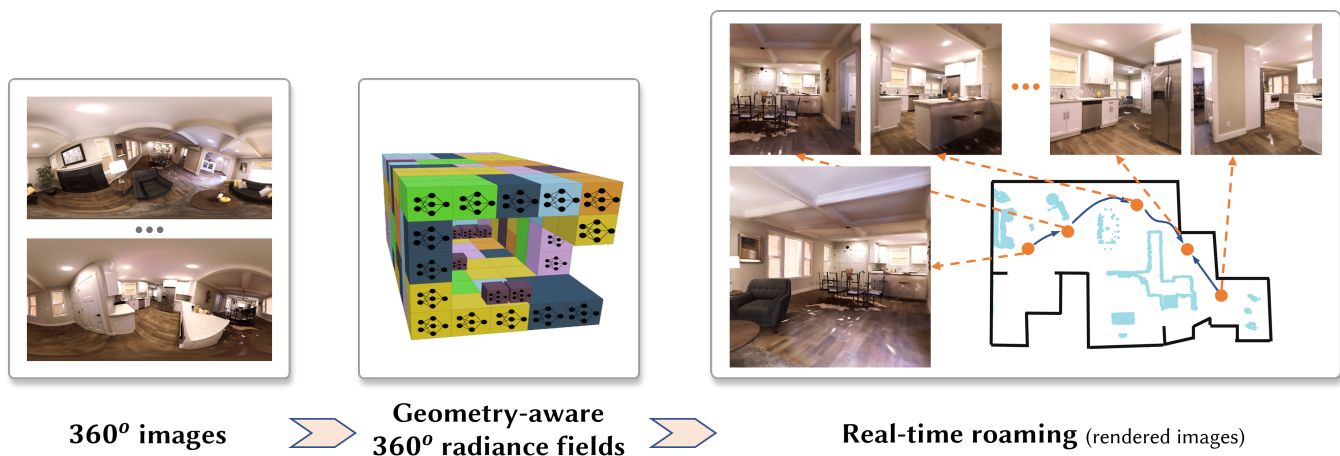


Figure 1: We propose a system for immersive 6-DoF roaming in the indoor scenarios. It takes 360° images as input to learn geometry-aware omnidirectional radiance fields which consist of plentiful adaptive-assigned neural perceptrons (illustrated as different colors in the middle figure). It can render novel-view images at real-time speed on a single GPU GTX1080. The system also outputs the floorplan of the observed scene to enhance immersion for VR applications.

ABSTRACT

Neural radiance field (NeRF) has recently achieved impressive results in novel view synthesis. However, previous works on NeRF mainly focus on object-centric scenarios. They would suffer observable performance degradation in outward-facing and large-scale scenes due to limiting positional encoding capacity. To narrow the gap, we explore radiance fields in a geometry-aware fashion. We estimate explicit geometry from the omnidirectional neural radiance field that was learned from multiple 360° images. Relying on the recovered geometry, we use an adaptive divide-and-conquer strategy

to slim and fine-tune the radiance fields and further improve render speed and quality. Quantitative and qualitative comparisons among baselines illustrated our predominant performance in large-scale indoor scenes and our system supports real-time VR roaming.

CCS CONCEPTS

• Computing methodologies → Rendering; Virtual reality.

KEYWORDS

Novel view synthesis, Neural rendering, Photorealistic imagery

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1 INTRODUCTION

Recently neural rendering has attracted great attention and demonstrated impressive rendering quality. Such learned view synthesis methods exploit neural networks to implicitly represent the structure and appearance of captured objects. The neural radiance field, pioneered by NeRF [Mildenhall et al. 2020] is currently the most promising path. NeRF utilizes volume rendering techniques and multi-layer perceptrons to regress density and view-dependent color per ray. By densely sampling along each ray, it can generate photorealistic results even when encountered with transparent objects and thin structures. The following NeRF-based methods [Baron et al. 2021; Pumarola et al. 2021] have made great efforts and modifications to increase rendering quality and speed. However, most of them focus on object-centric or small-scale scenes where placements of cameras are constrained during the training and inference phases. When we apply these methods to challenging situations of complex indoor scene synthesis, the rendering images are blur and lack high-frequency information due to insufficient representational capacity of positional encoders.

In this work, we target real-time performance that is essential for immersive applications. Therefore, instead of unitizing higher dimension of positional encoders and deeper neural networks, we decomposed the global radiance field into a set of local fields and apply tiny neural networks to tackle each field. We prove that proper decomposition of radiance fields can not only increase training speed but also better recover high-frequency details for fidelity rendering. In addition, considering that the consumer-level 360° camera is becoming more and more accessible, our system uses a 360° camera for data collection. A 360° neural radiance field (360NeRF) for the whole space is learned from all the input panoramas firstly. Second, we use 360NeRF to generate panoramic depth and uncertainty of the scene simultaneously, and then use a progressive probabilistic geometry estimation to reconstruct the occupancy map and floorplan of the scene, as Fig 2 shown. The geometric information effectively constrains the sampling range of the ray which allows us to adaptively slim a neural radiance field into numbers of sub-fields and fine-tune the radiance fields. The proposed pipeline can efficiently digitize real world for real-time 360° indoor-scene roaming that achieves 6-DoF flexibility.

2 BACKGROUND

NeRF [Mildenhall et al. 2020] represents the scene as a function of Cartesian coordinates, outputs densities and emitted radiance colors at those querying locations. In order to get the representation, for each pixel in the input images, a camera ray is marched from its camera center through the scene within the near and far sampling range $[t_n, t_f]$. For any 3D point $\mathbf{p}_j(x, y, z)$ in the ray, a global MLP model is applied to transform the concatenation of the point location (x, y, z) and viewing direction (represented by a unit vector $\mathbf{d}$) into a 1-dimension density (σ) and a 3-dimension RGB color (c). Since the density is only dependent to point position, the model decomposes the mapping into predicting the density using only the location as well as predicting the color using both the location and viewing direction. Specifically, the mapping is represented as a two-stage MLP. The first 8 fully-connected layers of the network, denoted as $MLP_{1:8}$, take as input the 3D position

$\mathbf{p}_j(x, y, z)$, and output the density σ and a 256-dimension radiance feature f_c . The combination of the feature f_c and viewing direction $\mathbf{d}$ is processed by the final fully connected layer MLP_9 , and decoded into the radiance $c(\mathbf{p}_j)$ in terms of RGB value. The MLP model can be represented as:

$$\begin{aligned}\sigma, f_c &= MLP_{1:8}(x, y, z) \\ c(\mathbf{p}_j) &= MLP_9(f_c, \mathbf{d}).\end{aligned}\quad (1)$$

To improve rendering results, it utilizes positional encoding (which is borrowed from the Transformer [Vaswani et al. 2017]) and maps the input coordinate vectors into a high-dimension space. The mapping is useful in modeling high-frequency functions. For each coordinate component x_i in both position and viewing direction, the D -dimension encoding is formulated as:

$$\mathcal{F}(x_i) = (\sin 2^0 \pi x_i, \cos 2^0 \pi x_i, \dots, \sin 2^{D-1} \pi x_i, \cos 2^{D-1} \pi x_i). \quad (2)$$

In addition, NeRF jointly optimizes two networks of same architecture as Eq. 1 with different parameters to increase rendering efficiency and avoid over-sampling in free spaces. A coarse network is evaluated on a set of N_c locations using stratified sampling. Based on the output densities of the coarse network, the coefficients which could be represented as a function of density are normalized as weights to form a piecewise-constant PDF (probability density function). Additional N_f points are sampled under this distribution. All $N_c + N_f$ points are used to query densities and colors through the fine network and render the final color of the ray.

3 METHOD

3.1 Omnidirectional Neural Radiance Field

The involved coordinate systems used in NeRF are all Cartesian while a panorama uses the spherical polar coordinate system. The camera ray cast from a panoramic pixel can be formulated as $\mathbf{r} = \mathbf{o} + r\mathbf{d}(\phi, \theta, 1)$, where $\mathbf{o}$ is the camera center, $\mathbf{d}$ is the ray direction and r is the radial distance of a point in the ray. ϕ, θ denote longitude and latitude of the spherical surface. Instead of transforming the coordinate to a 3D Cartesian coordinate, we conducts sampling in the spherical coordinate system, $\mathbf{p}_j(\phi, \theta, 1/r)$, and queries a MLP for corresponding color and density. This parameterization to model the omnidirectional neural radiance field, is referred as 360NeRF. Therefore, combining the initial MLP model (Eq. 1) and positional encoding (Eq. 2), our model could be rewritten as:

$$\begin{aligned}\sigma, f_c &= MLP_{1:8}(\mathcal{F}(\mathbf{p}_j)) \\ c(\mathbf{p}_j) &= MLP_9(f_c, \mathcal{F}(\mathbf{d})),\end{aligned}\quad (3)$$

where the positional embedding function $\mathcal{F}$ is applied individually on each component of a vector.

Finally, we accumulate the densities and radiance of N samplings along the ray $\mathcal{R}_i$ to get the final color $\hat{C}(\mathcal{R}_i)$ and depth $\hat{D}(\mathcal{R}_i)$ estimation of the pixel, similar to the volume rendering formulated in NeRF. Supposing the sampling position along the ray is t_i , the corresponding color and depth can be calculated by :

$$\begin{aligned}\hat{C}(\mathcal{R}_i) &= \sum_{i=1}^N T_i c_i \\ \hat{D}(\mathcal{R}_i) &= \sum_{i=1}^N T_i t_i,\end{aligned}\quad (4)$$

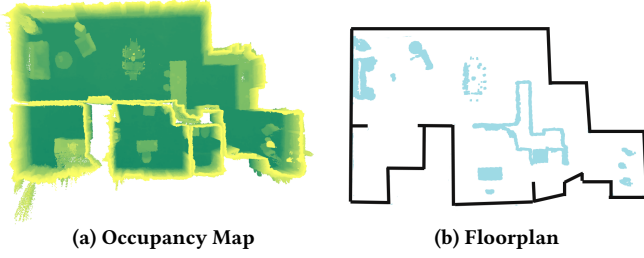


Figure 2: The occupancy map and floorplan are recovered from neural radiance field by progressive probabilistic geometry estimation algorithms

where the weight

$$T_i = \exp \left(- \sum_{k=1}^{i-1} \sigma_k (t_{k+1} - t_k) \right) (1 - \exp(-\sigma_i (t_{i+1} - t_i))).$$

3.2 Progressive Probabilistic Geometry Estimation

In structured scenes, there are a lot of open spaces and the distributions of objects are spatially sparse. It means that the density of radiance fields is highly imbalanced. If we uniformly sample along the rays, it is highly inefficient and easy to get trapped in the local minimum. Conversely, by skipping empty spaces and upsampling, occupied volumes are able to accelerate volume rendering and effectively improve the rendering quality.

3.2.1 Occupancy Map. To extract geometry from NeRF, we formulated occupancy recovery as a progressive probability estimation problem. Intuitively, the present state of each voxel depends on the current measurement and the previous estimation. The probabilistic model [Yguel et al. 2008] could be formulated in logit notation:

$$\text{logit}(p|z_{1:t}) = \text{logit}(p|z_{1:t-1}) + \text{logit}(p|z_t), \quad (5)$$

where $\text{logit}(p) = \log \frac{p}{1-p}$ and p is the probability of occupancy; $z_{1:t}$ represents the accumulated observation from the start to time t and z_t is the current observations at time t ; With the incoming observations, it can progressively update the occupied probability of each voxel. The range of $\text{logit}(p)$ is from -2 to 3.5 which indicates the occupied probability from 0.12 to 0.97. When the probability is larger than the threshold, the voxel will be considered occupied. We go through locations in the training set which is generally sufficient to reconstruct an appropriate occupancy map of the scene (see Fig. 2a). This process is depicted in the first part of Algorithm 1.

3.2.2 Floorplan. Given the occupancy map, we also estimate the floorplan consisting of the inner surface of the wall and any objects in the scene that obstruct the navigation path (see Fig. 2b). To better filter out noisy voxels from the occupancy map, we project a range of occupied voxels in the upper slicing regions (60% to 65% of the height to avoid head jamb for multi-room contours) to a floor-parallel plane to form an initial layout. To determine the slicing direction which is floor-parallel, we need to estimate the direction of the gravity of the reconstructed scene. Benefit from 360° images, we can get the gravity direction of the structured scene via calculating three vanishing points [Sun et al. 2021]. After

ALGORITHM 1:

Density distribution estimation and networks assignment

Input: 360NeRF and the amount of networks n
 /* Stage I */
 Sample positions: $Q \leftarrow$ training set;
for $q \in Q$ **do**
 distance map, uncertainty map $\leftarrow$ 360NeRF (q);
 Updating occupied probability map using Eq.5;
end
 Obtained occupancy map O and the amount of occupied voxel S ;
 /* Stage II */
 Size of each sub-field $s \leftarrow S/n$;
for $k \leftarrow 1$ **to** n **do**
 Pop an unassigned voxel with minimum coordinate from O as the anchor to search $s - 1$ nearest unassigned neighbors;
 Assign these s new voxels into k^{th} network;
end

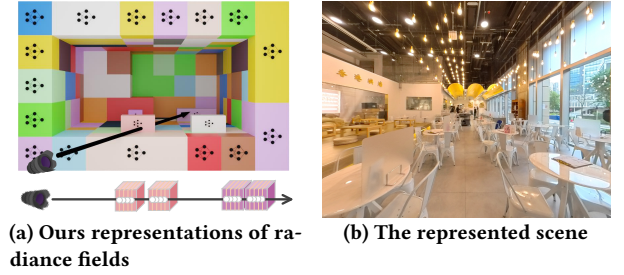


Figure 3: We use the occupancy map to adaptively decompose the 360NeRF into n sub-fields and assign the tiny MLP with the same network structure to handle different regions. Each sub-field has a similar amount of occupied voxels. The actual number depends on the scene geometry in practice. For the illustration purpose, each sub-field corresponds to two occupied voxels. The resultant geometry-aware field allows effective empty space skipping and dense sampling in occupied voxels to improve rendering quality.

aligning the 360° images with gravity, we are able to estimate the height and extract the floor-parallel plane.

3.3 Adaptive Radiance Fields Slimming and Fine-tuning

In a large-scale scene, the distribution of objects and their corresponding radiance fields are highly imbalanced. As the empty space does not contribute to volume rendering, we should avoid sampling and wasting the capacity of neural perceptron in empty spaces. Based on the extracted occupancy map, we can obtain the density distribution of the radiance fields which can be represented by the number of the occupied voxels. We then decompose the field into n sub-fields with a similar amount of occupied voxels as described in the second stage of Algorithm 1. This step only requires one hyperparameter n , the number of tiny MLPs, which can be adjusted according to computational resources and demand

Table 1: Comparison of Novel View Synthesis. Our system contains 2048 tiny MLPs and achieves the best performance. Approximate time to render one image is also reported.

Scene		NeRF	Mip-NeRF	KiloNeRF	Ours
Real Scenes	PSNR $\uparrow$	22.443	22.530	21.426	24.679
	SSIM $\uparrow$	0.672	0.670	0.690	0.755
	LPIPS $\downarrow$	0.339	0.336	0.283	0.209
Synthesis	PSNR $\uparrow$	34.018	33.568	36.229	36.284
	SSIM $\uparrow$	0.931	0.925	0.951	0.955
	LPIPS $\downarrow$	0.096	0.103	0.058	0.046
Inference Speed (s)		60	60	0.04	0.03

in terms of rendering performance. After this, each of the n tiny MLPs is used to cover each sub-field which is illustrated in the Fig.3. Eventually, our system can make use of the capability of neural networks and reduce the usage of MLPs while maintaining high-fidelity rendering. The initial weights of tiny MLPs are distilled from the 360NeRF to increase the fine-tuning speed. Similar to distillation procedures [Reiser et al. 2021], each MLP $\mathcal{M}_k$ will sample I corresponding points in the 360NeRF to obtain the global radiance values c_i and densities σ_i . $\mathcal{M}_k$'s parameters are optimized via minimizing the mean squared errors between global and local values (c_i^k, σ_i^k) , which is formulated as:

$$\mathcal{L}_k = \frac{1}{I} \sum_{i \in I} \|\sigma_i^k - \sigma_i\|_2^2 + \|c_i^k - c_i\|_2^2. \quad (6)$$

After initialization, the parameters of all tiny neural networks will be fine-tuned by using the original training images. During the fine-tuning phase, it will densely sample the occupied voxels instead of uniform sampling along the rays. As radiance fields are slimmed and a vast amount of empty sampling is being avoided, the fine-tuning process can be converged quickly.

4 RESULTS

We evaluated the novel view synthesis performance for both real and synthetic scenes quantitatively and qualitatively. Our proposed method is implemented in Pytorch and CUDA. The coarse training of a 360NeRF takes about 15 hours on 4 RTX 2080 Ti GPUs while the fine-tuning of thousands of tiny networks generally takes 4 hours after slimming. The average metrics on real and synthetic sets are reported in Table. 1. From the results reported in Table. 1, our complete system achieves the best quality on real and synthetic data while the render speed is two thousand times faster than NeRF [Mildenhall et al. 2020] and Mip-NeRF [Barron et al. 2021]. KiloNeRF [Reiser et al. 2021] using uniform decomposition can achieve comparable performance on synthetic scenes. But the performance of uniform decomposition is significantly degraded on real scenes. It is noticeable that results of real scenes generally have a worse performance compared to synthetic scenes. Since real scenes suffer from nonuniform lighting exposure cross views and camera pose approximation, while synthetic scenes provide ground-truth camera parameters and are rendered under the same illumination settings. Fig. 4 shows the a roaming example with long trajectories in LAB.

5 DISCUSSION AND CONCLUSION

We demonstrated how the system effectively extends the conventional NeRF into the geometry-aware radiance fields to generate high-quality 360° images for real-time indoor roaming. We also conduct an additional comparison with a typical way which uses 3D reconstruction [Schönberger and Frahm 2016; Schönberger et al. 2016] and texture mapping [Wachter et al. 2014] techniques to recover meshes for real scene roaming. As the COLMAP does not support 360° images, we crop the training panoramas into cube maps and then use the same camera poses to run the COLMAP. An illustrative example is shown in Fig.5. Our approach produces real-time photorealistic renderings and has a dramatic advantage over conventional 3D reconstruction which shows obvious geometry and texture artifacts. 360Roam has a large potential for VR roaming for large-scale indoor real scenes with less resource requirements and manpower.

**Figure 4: Roaming in the scene LAB using perspective view.**



(a) COLMAP



(b) Ours

Figure 5: Comparison to 3D reconstruction and texturing. We show a perspective result of typical 3D reconstruction (COLMAP [Schönberger and Frahm 2016]) and texture mapping (MVS-Texturing [Waechter et al. 2014]) on the left, and corresponding rendered view from our system on the right. Reconstructed textured model has obvious geometry and texture artifacts and leads to non-photorealism, while ours is much more photorealistic and suitable for VR roaming.

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